

Teaching Statement

I have had the pleasure of teaching mathematics continuously for the past 14 years. Throughout that time, I have found that my research and teaching lives have helped to balance one another by offering an opportunity to focus on familiar mathematical material and providing intellectual humility to empathize with my students as they grapple with learning unfamiliar material. Below I will comment on 3 important components of my teaching:

1. Provide well motivated and straightforward introductions to course content, focusing on how new material is connected to concepts students already understand,
2. Curate opportunities for students to interact and struggle with new material in order to facilitate mastery,
3. Create consistent and regular opportunities for bidirectional course feedback between instructor and students.

Introducing material: In my experience, how material is introduced to students has considerable impact on their subsequent understanding. In a vacuum, mathematical concepts, definitions, and notations can seem mysterious, unmotivated, and unintuitive. This is somewhat ironic, given that mathematical ideas typically emerge from an attempt to solve a specific problem. As a result, I take the approach that context is just as important as content. In practice, I strive to provide ample context (both mathematical and historical) and examples when introducing new material to students. For instance, when trying to motivate the integral formula for the length of a parameterized curve, I will begin by iteratively approximating the length of the curve using more and more line segments and show how this naturally leads to the integral formula.

Learning activities: When I first began teaching, I believed that if I crafted my lectures just right that all of my students would be able to learn the material with ease. However, slowly I began to see that at a certain point there were decreasing returns for putting more time into polishing my lectures. Over time I began to realize that rather than being viewed as a mechanism for learning, I should view teaching as a catalyst for learning. As a result, I have become more focused on helping to create opportunities for my students to learn course material. One of my favorite methods is to create worksheets for my students to work on in class in small groups. These worksheets typically consist of 3-5 questions of increasing difficulty and I try to focus on qualitative questions rather than quantitative ones. While the students work on the questions in groups I circulate and eavesdrop on their progress, and try to intervene only if students seem quite stuck. I find that these sessions often help the students realize where some of their confusion lies and become more confident in their knowledge as they are able to collaboratively solve problems. In addition to my own positive experience with these sessions, I regularly have students comment positively on them in end of semester surveys.

Feedback: As learning and teaching are iterative processes, it is crucial to have consistent and meaningful feedback between instructor and students. There are the obvious opportunities to provide feedback to students (e.g. exams, homework assignments, etc.), but these

high stakes events are typically infrequent and since these events heavily influence students' grades, this feedback often comes after it is too late to make changes. In response, I try to incorporate many low/no stakes opportunities for student feedback in my courses. The worksheet problems mentioned in the previous section are a good example of this type of no stakes feedback. This type of feedback happens almost every day in class, so students are getting real time feedback of their understanding with enough time to make corrections before they are expected to complete a high stakes assessment.

I also believe it is important that this feedback be bidirectional. While it is crucial for the student to receive feedback from the instructor, it is also important for students to provide feedback to the instructor about how the course is proceeding. Each semester, I provide an anonymous mid-semester feedback survey to my students. In this survey I ask the students questions about what is going well and what can be improved, as well as self assessment questions about how they have been preparing for the class. On several occasions I have received feedback that has allowed me to make small changes to the course that have had big impacts. For instance, in the past I have been asked to give a couple more minutes on weekly quizzes and been made aware that my office hours were at a time when many students had a lab course. In both cases, this type of feedback from students helped to improve the course delivery in meaningful ways.

Conclusion: These three components form the foundation of my teaching philosophy. Through them I attempt to provide meaningful instruction to students in a class environment that is conducive to learning.